Model 1: Abundance Plots

This file contains the processing code for Model 1: Comparing Temporally Blind and Temporally Explicit Metrics. Before running this file, you must have extinction time and population size outputs from running Model 1 for both TPCs using Model1.nb; edit import and export locations (via ‘saveto’) for your own device. Use to generate Fig 4. **(*7 Nov 2025 update:*** Corrected typo in beta param of TPC2 and updated color scheme + line thickness for Fig 4 (plus adjusted max/min bounds to match between species for easier comparison); fixed glitch in CD index outputs -- CDMatrixT was using *p* instead of *pT* (though graph in manuscript was already corrected from a separate file) ***)**

```
saveto = "/SAVETO/";
```

Load Data

Run import code before creating any figures. Edit import/export file names/locations for your local system and, if necessary, edit loaded parameter space to match your imported run. Default size is tmax2000_dims21,4,1,1000, corresponding with 1000 runs of 2000 time steps at each of 21 autocorrelation levels, 4 mean temperatures, and 1 standard deviation of temperature.

```
In[ ]:= extTime = Import[saveto <> "extMetrics8_extTime_tmax2000_dims21,4,1,1000.m", "MX"];
p = Import[saveto <> "extMetrics8_popsize_tmax2000_dims21,4,1,1000.m", "MX"];

extTimeT =
  Import[saveto <> "extMetrics8_trop_extTime_tmax2000_dims21,4,1,1000.m", "MX"];
pT = Import[saveto <> "extMetrics8_trop_popsize_tmax2000_dims21,4,1,1000.m", "MX"];

tmax = 2000; (*2000*)
maxsims = 1000; (*1000*)
γvals = Range[0, 2, .1]; (*21*)
μTvals = Range[23, 26, 1]; (*4*)
σTvals = Range[3, 3, 1]; (*1*)
```

Calculate Temporally Blind Metrics

Calculate the expected population abundances given the means and standard deviation of temperature without any input from the simulations.

```
In[ ]:= (*TEMPORATE - TPC1*)
gain[T_] := Exp[-((T - TI)^2) / β];
loss[T_] := ma Exp[mb T] + mc;
```

```
r[T_] := (gain[T] - loss[T]) / .755;

(* Parameters of the TPC model *)
TI = 29;
β = 180;
ma = 0.000005;
mb = 0.4;
mc = 0.05;

α = .001;

(*array of temperatures for each parameter set*)
W = ConstantArray[0, {Length[μTvals], Length[σTvals], tmax}];
Do[W⟦i, j, k⟧ = InverseCDF[NormalDistribution[μTvals⟦i⟧, σTvals⟦j⟧], k * (1 / tmax)],
 {i, 1, Length[μTvals]}, {j, 1, Length[σTvals]}, {k, 1, tmax - 1}]
Do[W⟦i, j, tmax⟧ =
   InverseCDF[NormalDistribution[μTvals⟦i⟧, σTvals⟦j⟧], (tmax - .5) * (1 / tmax)],
  {i, 1, Length[μTvals]}, {j, 1, Length[σTvals]}];

(*set initial pop size to mean pop size for each parameter set*)
ninit = ConstantArray[0, {Length[μTvals], Length[σTvals]}];
Do[ninit⟦i, k⟧ = (Mean[Table[r[W⟦i, k, j⟧], {j, 1, tmax}]]) / α,
 {k, 1, Length[σTvals]}, {i, 1, Length[μTvals]}]
N[ninit]

l23 = ConstantArray[ninit⟦1⟧, Length[γvals]];
l23t = Transpose[{γvals, Flatten[l23]}];
l24 = ConstantArray[ninit⟦2⟧, Length[γvals]];
l24t = Transpose[{γvals, Flatten[l24]}];
l25 = ConstantArray[ninit⟦3⟧, Length[γvals]];
l25t = Transpose[{γvals, Flatten[l25]}];
l26 = ConstantArray[ninit⟦4⟧, Length[γvals]];
l26t = Transpose[{γvals, Flatten[l26]}];

(*TROPICAL - TPC2*)
gain[T_] := Exp[-((T - TI) ^ 2) / β];
loss[T_] := ma Exp[mb T] + mc;
r[T_] := (gain[T - 2] - loss[T - 2]) / .4067;

(* Parameters of the TPC model *)
TI = 35;
β = 170;
ma = 0.000005;
mb = 0.4;
```

```
mc = 0.05;

α = .001;

(*array of temperatures for each parameter set*)
W = ConstantArray[0, {Length[μTvals], Length[σTvals], tmax}];
Do[W⟦i, j, k⟧ = InverseCDF[NormalDistribution[μTvals⟦i⟧, σTvals⟦j⟧], k * (1 / tmax)],
 {i, 1, Length[μTvals]}, {j, 1, Length[σTvals]}, {k, 1, tmax - 1}]
Do[W⟦i, j, tmax⟧ =
   InverseCDF[NormalDistribution[μTvals⟦i⟧, σTvals⟦j⟧], (tmax - .5) * (1 / tmax)],
  {i, 1, Length[μTvals]}, {j, 1, Length[σTvals]}];

(*set initial pop size to mean pop size for each parameter set*)
ninit = ConstantArray[0, {Length[μTvals], Length[σTvals]}];
Do[ninit⟦i, k⟧ = (Mean[Table[r[W⟦i, k, j⟧], {j, 1, tmax}]]) / α,
 {k, 1, Length[σTvals]}, {i, 1, Length[μTvals]}]

N[ninit]

lt23 = ConstantArray[ninit⟦1⟧, Length[γvals]];
lt23t = Transpose[{γvals, Flatten[lt23]}];
lt24 = ConstantArray[ninit⟦2⟧, Length[γvals]];
lt24t = Transpose[{γvals, Flatten[lt24]}];
lt25 = ConstantArray[ninit⟦3⟧, Length[γvals]];
lt25t = Transpose[{γvals, Flatten[lt25]}];
lt26 = ConstantArray[ninit⟦4⟧, Length[γvals]];
lt26t = Transpose[{γvals, Flatten[lt26]}];
```

Out[ ]=

```
{{851.715}, {845.332}, {797.846}, {691.715}}
```

Out[ ]=

```
{{589.123}, {660.979}, {713.579}, {731.579}}
```

Process Simulation Data

Given the imported extinction times and population sizes, calculate the extinction metrics (abundance, stability, and probability of extinction) and create plotting arrays.

```
(*initialize functions for calculating the proportion variance index (PV)
 and consecutive disparity index (CD); only the latter is computed here*)
CDindex[data_List] := Module[{n, diffs}, n = Length[data];
  diffs = Abs[Log[data⟦2 ;;⟧ / data⟦1 ;; -2⟧]];
  Mean[diffs]]
(*PVindex[data_List]:=Module[{n,pairs,zsum},n=Length[data];
```

```
  pairs=Subsets[data,{2}];
  zsum=Total[(1-(Min[#]/Max[#]))&/@pairs];
  (2*zsum)/(n*(n-1))]*)

(*TEMPERATE*)
meansMatrix = ConstantArray[0, {Length[γvals], Length[μTvals], maxsims}];
stdevsMatrix = ConstantArray[0, {Length[γvals], Length[μTvals], maxsims}];
stabilityMatrix = ConstantArray[0, {Length[γvals], Length[μTvals], maxsims}];
extMatrix = ConstantArray[0, {Length[γvals], Length[μTvals], maxsims}];
cdMatrix = ConstantArray[0, {Length[γvals], Length[μTvals], maxsims}];

(*averaging over all timesteps before extinction within a simulation*)
Do[If[extTime⟦i, j, 1, l⟧ < tmax, extMatrix⟦i, j, l⟧ = 1, extMatrix⟦i, j, l⟧ = 0];
 meansMatrix⟦i, j, l⟧ = Mean[p⟦i, j, 1, l, 2 ;; extTime⟦i, j, 1, l⟧⟧];
 stdevsMatrix⟦i, j, l⟧ =
  StandardDeviation[p⟦i, j, 1, l, 2 ;; extTime⟦i, j, 1, l⟧⟧];
 stabilityMatrix⟦i, j, l⟧ = meansMatrix⟦i, j, l⟧ / stdevsMatrix⟦i, j, l⟧,
 {l, 1, maxsims}, {j, 1, Length[μTvals]}, {i, 1, Length[γvals]}]

Do[cdMatrix⟦i, j, l⟧ = CDindex[p⟦i, j, 1, l, 2 ;; extTime⟦i, j, 1, l⟧⟧ + .1],
 {l, 1, maxsims}, {j, 1, Length[μTvals]}, {i, 1, Length[γvals]}]
cdMatrix2 = ConstantArray[0, {Length[γvals], Length[μTvals]}];
Do[cdMatrix2⟦i, j⟧ = DeleteAnomalies[cdMatrix⟦i, j⟧],
 {j, 1, Length[μTvals]}, {i, 1, Length[γvals]}]

(*averaging across simulations with the same params*)
meansVals = ConstantArray[0, {Length[γvals], Length[μTvals]}];
meansQ1 = ConstantArray[0, {Length[γvals], Length[μTvals]}];
meansQ3 = ConstantArray[0, {Length[γvals], Length[μTvals]}];
stdevsVals = ConstantArray[0, {Length[γvals], Length[μTvals]}];
stabilityVals = ConstantArray[0, {Length[γvals], Length[μTvals]}];
propExt = ConstantArray[0, {Length[γvals], Length[μTvals]}];
cdVals = ConstantArray[0, {Length[γvals], Length[μTvals]}];
cdQ1 = ConstantArray[0, {Length[γvals], Length[μTvals]}];
cdQ3 = ConstantArray[0, {Length[γvals], Length[μTvals]}];

Do[meansVals⟦i, j⟧ = Mean[meansMatrix⟦i, j⟧];
 meansQ1⟦i, j⟧ = Quantile[meansMatrix⟦i, j⟧, .1];
 meansQ3⟦i, j⟧ = Quantile[meansMatrix⟦i, j⟧, .9];
 stdevsVals⟦i, j⟧ = Mean[stdevsMatrix⟦i, j⟧];
 stabilityVals⟦i, j⟧ = Mean[stabilityMatrix⟦i, j⟧];
 propExt⟦i, j⟧ = Total[extMatrix⟦i, j⟧ / maxsims];
 cdVals⟦i, j⟧ = Mean[cdMatrix2⟦i, j⟧];
 cdQ1⟦i, j⟧ = Quantile[cdMatrix2⟦i, j⟧, .1];
```

```
  cdQ3⟦i, j⟧ = Quantile[cdMatrix2⟦i, j⟧, .9],
  {j, 1, Length[μTvals]}, {i, 1, Length[γvals]}]

(*TROPICAL*)
meansMatrixT = ConstantArray[0, {Length[γvals], Length[μTvals], maxsims}];
stdevsMatrixT = ConstantArray[0, {Length[γvals], Length[μTvals], maxsims}];
stabilityMatrixT = ConstantArray[0, {Length[γvals], Length[μTvals], maxsims}];
extMatrixT = ConstantArray[0, {Length[γvals], Length[μTvals], maxsims}];
cdMatrixT = ConstantArray[0, {Length[γvals], Length[μTvals], maxsims}];

(*averaging over all timesteps before extinction within a simulation*)
Do[If[extTimeT⟦i, j, 1, l⟧ < tmax, extMatrixT⟦i, j, l⟧ = 1, extMatrixT⟦i, j, l⟧ = 0];
  meansMatrixT⟦i, j, l⟧ = Mean[pT⟦i, j, 1, l, 2 ;; extTimeT⟦i, j, 1, l⟧⟧];
  stdevsMatrixT⟦i, j, l⟧ =
   StandardDeviation[pT⟦i, j, 1, l, 2 ;; extTimeT⟦i, j, 1, l⟧⟧];
  stabilityMatrixT⟦i, j, l⟧ = meansMatrixT⟦i, j, l⟧ / stdevsMatrixT⟦i, j, l⟧;,
  {l, 1, maxsims}, {j, 1, Length[μTvals]}, {i, 1, Length[γvals]}]

Do[cdMatrixT⟦i, j, l⟧ = CDindex[pT⟦i, j, 1, l, 2 ;; extTimeT⟦i, j, 1, l⟧⟧ + .1],
  {l, 1, maxsims}, {j, 1, Length[μTvals]}, {i, 1, Length[γvals]}]
cdMatrixT2 = ConstantArray[0, {Length[γvals], Length[μTvals]}];
Do[cdMatrixT2⟦i, j⟧ = DeleteAnomalies[cdMatrixT⟦i, j⟧],
  {j, 1, Length[μTvals]}, {i, 1, Length[γvals]}]

(*averaging across simulations with the same params*)
meansValsT = ConstantArray[0, {Length[γvals], Length[μTvals]}];
meansQ1T = ConstantArray[0, {Length[γvals], Length[μTvals]}];
meansQ3T = ConstantArray[0, {Length[γvals], Length[μTvals]}];
stdevsValsT = ConstantArray[0, {Length[γvals], Length[μTvals]}];
stabilityValsT = ConstantArray[0, {Length[γvals], Length[μTvals]}];
propExtT = ConstantArray[0, {Length[γvals], Length[μTvals]}];
cdValsT = ConstantArray[0, {Length[γvals], Length[μTvals]}];
cdQ1T = ConstantArray[0, {Length[γvals], Length[μTvals]}];
cdQ3T = ConstantArray[0, {Length[γvals], Length[μTvals]}];

Do[meansValsT⟦i, j⟧ = Mean[meansMatrixT⟦i, j⟧];
  meansQ1T⟦i, j⟧ = Quantile[meansMatrixT⟦i, j⟧, .1];
  meansQ3T⟦i, j⟧ = Quantile[meansMatrixT⟦i, j⟧, .9];
  stdevsValsT⟦i, j⟧ = Mean[stdevsMatrixT⟦i, j⟧];
  stabilityValsT⟦i, j⟧ = Mean[stabilityMatrixT⟦i, j⟧];
  propExtT⟦i, j⟧ = Total[extMatrixT⟦i, j⟧ / maxsims];
  cdValsT⟦i, j⟧ = Mean[cdMatrixT2⟦i, j⟧];
  cdQ1T⟦i, j⟧ = Quantile[cdMatrixT2⟦i, j⟧, .1];
  cdQ3T⟦i, j⟧ = Quantile[cdMatrixT2⟦i, j⟧, .9],
```

```
 {j, 1, Length[μTvals]}, {i, 1, Length[γvals]}]

(*create plotting arrays*)
means23 = Transpose[{γvals, meansVals⟦All, 1⟧}];
means24 = Transpose[{γvals, meansVals⟦All, 2⟧}];
means25 = Transpose[{γvals, meansVals⟦All, 3⟧}];
means26 = Transpose[{γvals, meansVals⟦All, 4⟧}];

means23Q1 = Transpose[{γvals, meansQ1⟦All, 1⟧}];
means23Q3 = Transpose[{γvals, meansQ3⟦All, 1⟧}];
means24Q1 = Transpose[{γvals, meansQ1⟦All, 2⟧}];
means24Q3 = Transpose[{γvals, meansQ3⟦All, 2⟧}];
means25Q1 = Transpose[{γvals, meansQ1⟦All, 3⟧}];
means25Q3 = Transpose[{γvals, meansQ3⟦All, 3⟧}];
means26Q1 = Transpose[{γvals, meansQ1⟦All, 4⟧}];
means26Q3 = Transpose[{γvals, meansQ3⟦All, 4⟧}];

cd23 = Transpose[{γvals, cdVals⟦All, 1⟧}];
cd24 = Transpose[{γvals, cdVals⟦All, 2⟧}];
cd25 = Transpose[{γvals, cdVals⟦All, 3⟧}];
cd26 = Transpose[{γvals, cdVals⟦All, 4⟧}];

cd23Q1 = Transpose[{γvals, cdQ1⟦All, 1⟧}];
cd23Q3 = Transpose[{γvals, cdQ3⟦All, 1⟧}];
cd24Q1 = Transpose[{γvals, cdQ1⟦All, 2⟧}];
cd24Q3 = Transpose[{γvals, cdQ3⟦All, 2⟧}];
cd25Q1 = Transpose[{γvals, cdQ1⟦All, 3⟧}];
cd25Q3 = Transpose[{γvals, cdQ3⟦All, 3⟧}];
cd26Q1 = Transpose[{γvals, cdQ1⟦All, 4⟧}];
cd26Q3 = Transpose[{γvals, cdQ3⟦All, 4⟧}];

propExt23 = Transpose[{γvals, propExt⟦All, 1⟧}];
propExt24 = Transpose[{γvals, propExt⟦All, 2⟧}];
propExt25 = Transpose[{γvals, propExt⟦All, 3⟧}];
propExt26 = Transpose[{γvals, propExt⟦All, 4⟧}];

means23T = Transpose[{γvals, meansValsT⟦All, 1⟧}];
means24T = Transpose[{γvals, meansValsT⟦All, 2⟧}];
means25T = Transpose[{γvals, meansValsT⟦All, 3⟧}];
means26T = Transpose[{γvals, meansValsT⟦All, 4⟧}];

means23Q1T = Transpose[{γvals, meansQ1T⟦All, 1⟧}];
means23Q3T = Transpose[{γvals, meansQ3T⟦All, 1⟧}];
means24Q1T = Transpose[{γvals, meansQ1T⟦All, 2⟧}];
```

```
means24Q3T = Transpose[{γvals, meansQ3T[[All, 2]]}];
means25Q1T = Transpose[{γvals, meansQ1T[[All, 3]]}];
means25Q3T = Transpose[{γvals, meansQ3T[[All, 3]]}];
means26Q1T = Transpose[{γvals, meansQ1T[[All, 4]]}];
means26Q3T = Transpose[{γvals, meansQ3T[[All, 4]]}];

cd23T = Transpose[{γvals, cdValsT[[All, 1]]}];
cd24T = Transpose[{γvals, cdValsT[[All, 2]]}];
cd25T = Transpose[{γvals, cdValsT[[All, 3]]}];
cd26T = Transpose[{γvals, cdValsT[[All, 4]]}];

cd23Q1T = Transpose[{γvals, cdQ1T[[All, 1]]}];
cd23Q3T = Transpose[{γvals, cdQ3T[[All, 1]]}];
cd24Q1T = Transpose[{γvals, cdQ1T[[All, 2]]}];
cd24Q3T = Transpose[{γvals, cdQ3T[[All, 2]]}];
cd25Q1T = Transpose[{γvals, cdQ1T[[All, 3]]}];
cd25Q3T = Transpose[{γvals, cdQ3T[[All, 3]]}];
cd26Q1T = Transpose[{γvals, cdQ1T[[All, 4]]}];
cd26Q3T = Transpose[{γvals, cdQ3T[[All, 4]]}];

propExtT23 = Transpose[{γvals, propExtT[[All, 1]]}];
propExtT24 = Transpose[{γvals, propExtT[[All, 2]]}];
propExtT25 = Transpose[{γvals, propExtT[[All, 3]]}];
propExtT26 = Transpose[{γvals, propExtT[[All, 4]]}];
```

Create Individual Plots

Code for each individual subplot. Outputs repressed for processing time.

```
col1 = Lighter[Lighter[Red]];
col2 = Red;
col3 = Darker[Red];
col4 = Black;
wide = Thickness[0.01];

meansPlot = ListLinePlot[
  {means23, means23Q1, means23Q3, means24, means24Q1, means24Q3, means25,
   means25Q1, means25Q3, means26, means26Q1, means26Q3, l23t, l24t, l25t, l26t},
  PlotStyle → {{wide, col1}, None, None, {wide, col2}, None, None,
    {wide, col3}, None, None, {wide, col4}, None, None, {wide, Dashed, col1},
    {wide, Dashed, col2}, {wide, Dashed, col3}, {wide, Dashed, col4}},
  PlotRange → {{-0.02, 2.02}, {400, 900}}, FillingStyle → Opacity[0.1],
  Filling →
```

```
    {1 → {2}, 1 → {3}, 4 → {5}, 4 → {6}, 7 → {8}, 7 → {9}, 10 → {11}, 10 → {12}},
   Frame → {True, True, False, False}, FrameStyle → 14,
   FrameLabel → {None(*"Spectral Exponent γ"*), "Mean Population Abundance"},
   LabelStyle → Directive[Black], ImageSize → 500] ;(*subplot a: temp means*)

stabilityPlot = ListLinePlot[{cd26, cd26Q1, cd26Q3,
    cd25, cd25Q1, cd25Q3, cd24, cd24Q1, cd24Q3, cd23, cd23Q1, cd23Q3},
   PlotStyle → {{wide, col4}, None, None, {wide, col3}, None, None, {wide, col2},
     None, None, {wide, col1}, None, None}, FillingStyle → Opacity[0.1],
   Filling → {10 → {11}, 10 → {12}, 7 → {8}, 7 → {9}, 4 → {5}, 4 → {6}, 1 → {2}, 1 → {3}},
   FillingStyle → Opacity[0.1], PlotRange → {{-0.02, 2.02}, {-.01, .55}},
   Frame → {True, True, False, False}, FrameStyle → 14,
   FrameLabel → {(*"Spectral Exponent γ",*)None, "Consecutive Disparity Index"},
   LabelStyle → Directive[Black], ImageSize → 510];(*subplot b: stability*)

extinctionPlot = ListLinePlot[{propExt23, propExt24, propExt25, propExt26},
   PlotStyle → {{wide, col1}, {wide, col2}, {wide, col3}, {wide, col4}},
   PlotRange → {{-0.02, 2.02}, Automatic},
   Frame → {True, True, False, False}, FrameStyle → 14,
   FrameLabel → {"Spectral Exponent γ", "Probability of Extinction"},
   LabelStyle → Directive[Black], ImageSize → 500];
(*subplot c: temp extinction*)

meansPlotT = ListLinePlot[{lt23t, lt24t, lt25t, lt26t,
   means23T, means23Q1T, means23Q3T, means24T, means24Q1T, means24Q3T,
   means25T, means25Q1T, means25Q3T, means26T, means26Q1T, means26Q3T},
  PlotStyle → {{wide, Dashed, col1}, {wide, Dashed, col2}, {wide, Dashed, col3},
    {wide, Dashed, col4}, {wide, col1}, None, None, {wide, col2},
    None, None, {wide, col3}, None, None, {wide, col4}, None, None},
  PlotRange → {{-0.02, 2.02}, {400, 900}}, FillingStyle → Opacity[0.1],
  Filling →
   {5 → {6}, 5 → {7}, 8 → {9}, 8 → {10}, 11 → {12}, 11 → {13}, 14 → {15}, 14 → {16}},
  Frame → {True, True, False, False}, FrameStyle → 14,
  (*FrameLabel→{(*None,*)"Spectral Exponent γ","Mean Population Abundance"},*)
  LabelStyle → Directive[Black], ImageSize → 500](*subplot d: trop means*)

stabilityPlotT =
  ListLinePlot[{cd23T, cd23Q1T, cd23Q3T, cd24T, cd24Q1T, cd24Q3T, cd25T, cd25Q1T,
    cd25Q3T, cd26T, cd26Q1T, cd26Q3T}, PlotStyle → {{wide, col1}, None, None,
     {wide, col2}, None, None, {wide, col3}, None, None, {wide, col4}, None, None},
   PlotRange → {{-0.02, 2.02}, {-.01, .55}}, FillingStyle → Opacity[0.1],
   (*PlotLabel→Style["Specialist",16],*)
   Filling → {10 → {11}, 10 → {12}, 7 → {8}, 7 → {9}, 4 → {5}, 4 → {6}, 1 → {2}, 1 → {3}},
   Frame → {True, True, False, False}, FrameStyle → 14,
```

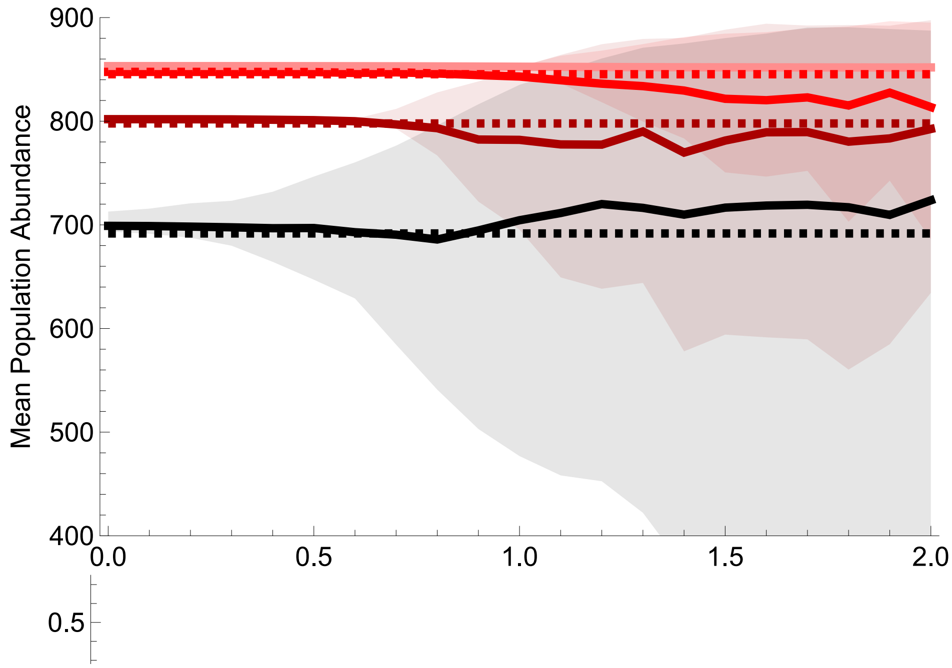
```
    (*FrameLabel→{None(*"Spectral Exponent γ"*),"Population Stability"},*)
    LabelStyle → Directive[Black], ImageSize → 510];(*subplot e: trop stability*)

extinctionPlotT = ListLinePlot[{propExtT23, propExtT24, propExtT25, propExtT26},
    PlotStyle → {{wide, col1}, {wide, col2}, {wide, col3}, {wide, col4}},
    PlotRange → {{-0.02, 2.02}, Automatic},
    Frame → {True, True, False, False}, FrameStyle → 14,
    FrameLabel → {"Spectral Exponent γ", None(*"Probability of Extinction"*)},
    LabelStyle → Directive[Black], ImageSize → 500];(*subplot f: trop extinction*)
```

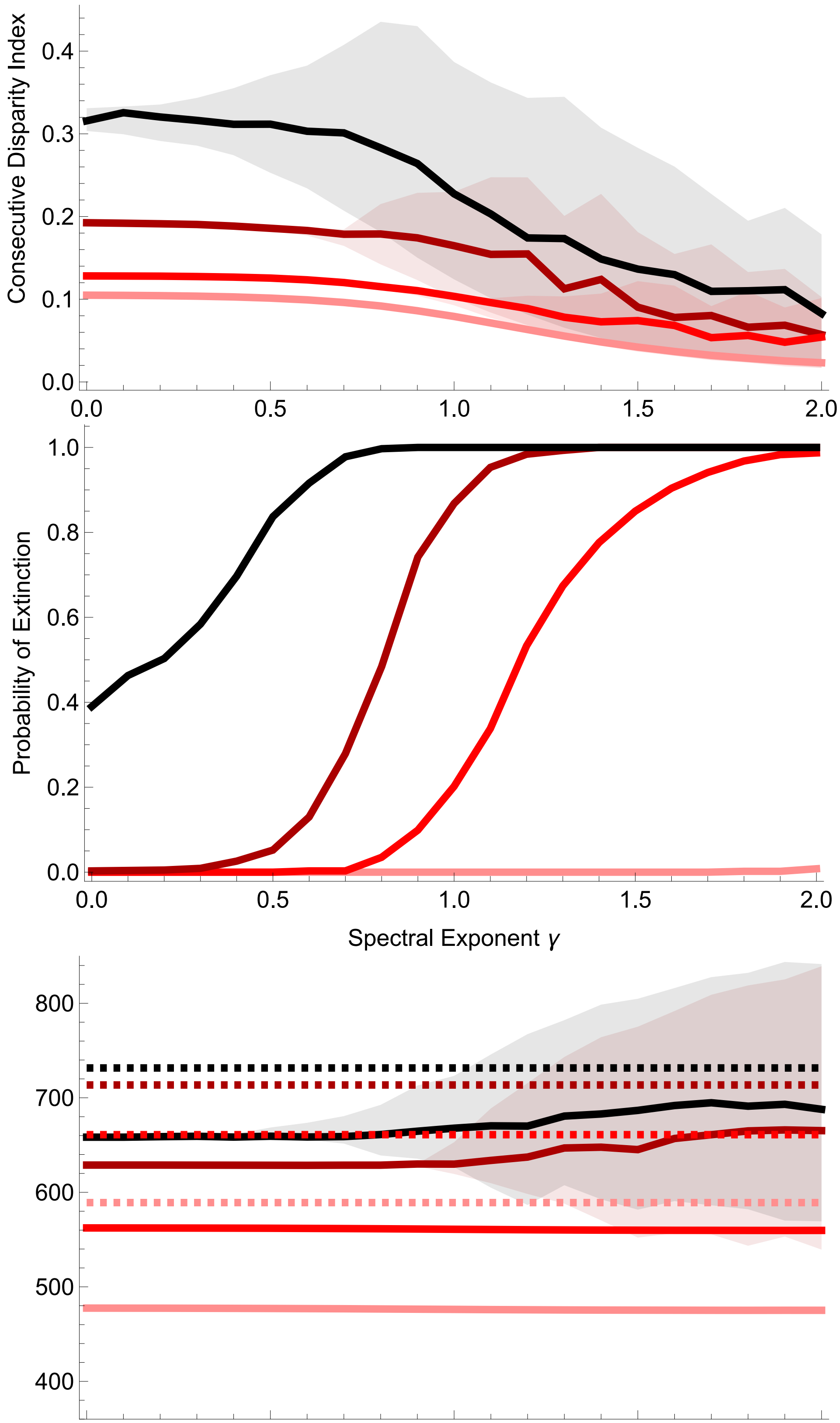
Fig 4: Extinction Metrics

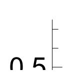
Create Fig 4. Labeling added in Adobe Illustrator 2024.

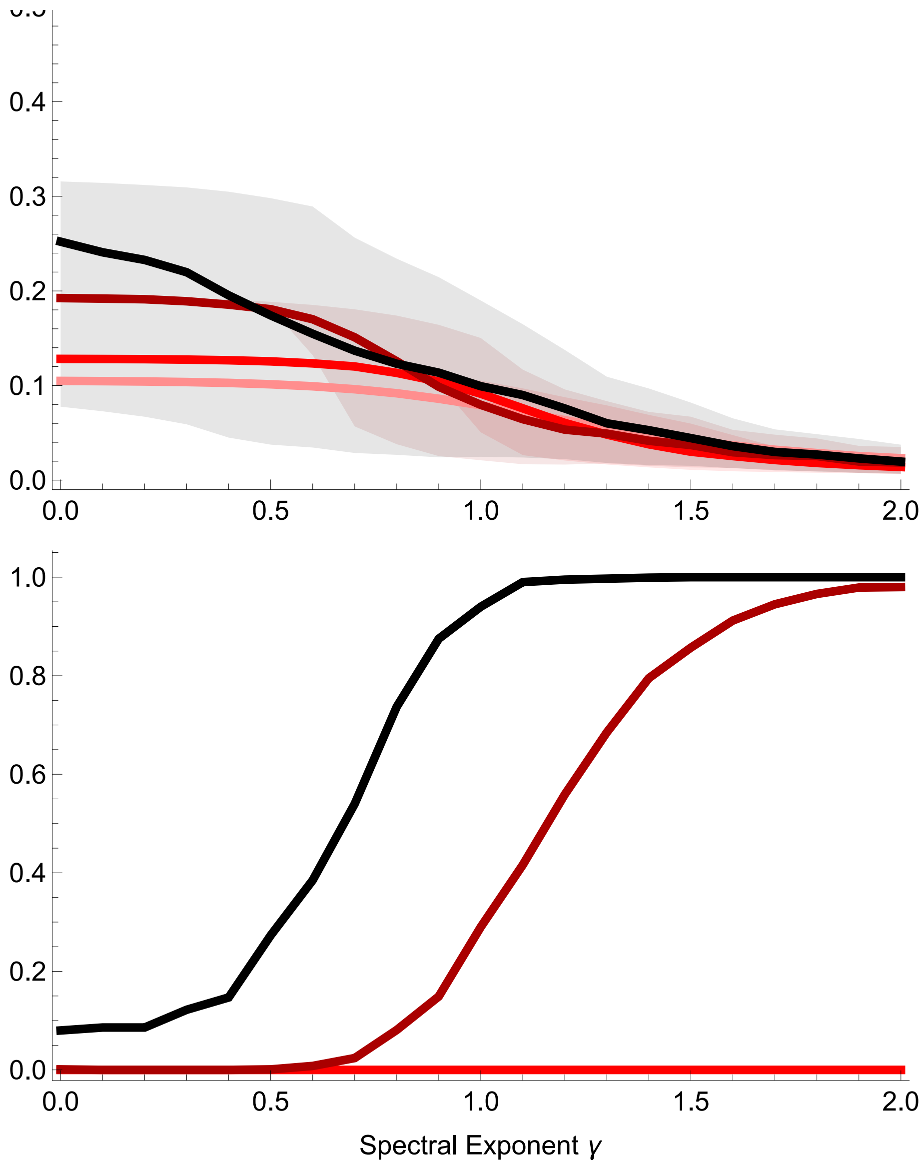
```
fig4 = GraphicsColumn[{meansPlot, stabilityPlot, extinctionPlot, meansPlotT,
     stabilityPlotT, extinctionPlotT}, Spacings → {0, 0}, ImageSize → 500];
(*fig4=GraphicsGrid[{{meansPlot,meansPlotT},{stabilityPlot,stabilityPlotT},
    {extinctionPlot,extinctionPlotT}},Spacings→{0,0},ImageSize→1000]*)

Export[saveto <> "Figure4.pdf", fig4, ImageResolution → 1000];
(*Export[saveto<>"meansPlot.pdf",meansPlot,ImageResolution→1000]
 Export[saveto<>"meansPlotT.pdf",meansPlotT,ImageResolution→1000]
 Export[saveto<>"stabilityPlot.pdf",stabilityPlot,ImageResolution→1000]
 Export[saveto<>"stabilityPlotT.pdf",stabilityPlotT,ImageResolution→1000]
 Export[saveto<>"extinctionPlot.pdf",extinctionPlot,ImageResolution→1000]
 Export[saveto<>"extinctionPlotT.pdf",extinctionPlotT,ImageResolution→1000]*)
```

Out[ ]=




Consecutive Disparity Index
0.4
0.3
0.2
0.1
0.0
0.0
0.5
1.0
1.5
2.0
Probability of Extinction
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.5
1.0
1.5
2.0
Spectral Exponent γ
800
700
600
500
400
0.0
0.5
1.0
1.5
2.0


0.5


0.4
0.3
0.2
0.1
0.0
0.0
0.5
1.0
1.5
2.0
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.5
1.0
1.5
2.0
Spectral Exponent γ